Midterm 2 practice 1

These are midterm practice problems, not for submission. On the actual midterm, some of the true/false questions will not ask for justification, but providing justification gives more practice.

Practice 1.1. For each statement in the list below: explain whether the statement is true or false, and prove your answer.

Wherever the symbol X appears, it denotes a metric space. Unless otherwise stated, assume the usual distance function on $\mathbb{R}$.

(1) Every bounded subset of $\mathbb{R}$ is contained in a compact set.

(2) A series $\sum_{n=1}^{\infty} x_n$ converges if and only if $x_n \to 0$.

(3) If $E \subset X$ and (x_n) is a sequence in E which converges in X, then (x_n) converges in E.

(4) If $E \subset X$ and (x_n) is a sequence in E, then (x_n) is Cauchy in E if and only if it is Cauchy in X.

(5) If (x_n) is a sequence in an a complete metric space with $d(x_{n+1}, x_n) \leq 2^{-n}$ then (x_n) converges.

(6) Given a sequence of real numbers (x_n), if $\liminf_{n\to\infty} x_n \neq \limsup_{n\to\infty} x_n$, then (x_n) is not Cauchy.

(7) If (x_n) is an unbounded sequence of real numbers, then there exists a subsequence (x_{n_k}) such that $|x_{n_k}| \to \infty$.

(8) If $\sum_{n=1}^{\infty} x_n$ converges, then for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $|\sum_{k=1}^{\infty} x_{k+n}| < \epsilon$ for all $n \geq N$.

(9) If X has the discrete metric ($d(x, y) = 1$ for $x \neq y$), then X is complete.

(10) If $f : \mathbb{R} \to \mathbb{R}$ is a function such that $\lim_{x\to a} f(x) = 0$ for all $a \in \mathbb{R}$, then $f(a) = 0$ for all $a \in \mathbb{R}$.

Practice 1.2. Let $\phi : \mathbb{N} \to \mathbb{N}$ be a bijection such that there exists $N \in \mathbb{N}$ such that $\phi(n) = n$ for all $n \geq N$. Prove that if $\sum_{n=1}^{\infty} x_n$ is any convergent series, then $\sum_{n=1}^{\infty} x_{\phi(n)}$ also converges and $\sum_{n=1}^{\infty} x_{\phi(n)} = \sum_{n=1}^{\infty} x_n$ (even if the series converges conditionally).

Practice 1.3. (1) Does the limit of the following function $\mathbb{R}^2 \backslash (0, 0)$ exist as $(x, y) \to (0, 0)$? If so what is it? Prove your answer.

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$$

(2) Is the following function continuous? Prove your answer.

$$f(x, y) = \begin{cases} \frac{(x^2-y^2)^2}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$